

YARENSK, Russian Federation

WMO: 227980

Lat: 62.1717N

Lon: 49.1094E

Elev: 92

StdP: 100.23

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.4	(c) -30.3	(d) -36.3	(e) 0.1	(f) -33.4	(g) -33.1	(h) 0.2	(i) -30.3	(j) 7.1	(k) -7.5	(l) 6.0	(m) -8.1	(n) 0.7	(o) 40	(p) 0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.6	(c) 27.9	(d) 18.7	(e) 25.8	(f) 17.5	(g) 23.9	(h) 16.7	(i) 20.1	(j) 25.5	(k) 18.9	(l) 23.8	(m) 17.8	(n) 22.0	(o) 2.4	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.3	13.3	22.3	17.2	12.4	20.6	16.1	11.5	20.0	58.0	25.4	54.1	23.7	50.6	22.0	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.6	5.8	5.2	-37.3	31.6	2.9	2.0	-39.4	33.0	-41.1	34.2	-42.7	35.3	-44.7	36.7
			-37.4	22.4	2.9	2.4	-39.5	24.1	-41.2	25.5	-42.8	26.9	-44.8	28.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-14.2	-10.6	-6.2	1.7	8.1	14.3	16.5	14.1	8.3	1.4	-5.9	-11.3
	DBStd	12.00	8.02	7.84	6.13	4.74	5.63	4.36	3.54	4.01	3.79	4.09	6.65	8.84
	HDD10.0	3691	751	577	502	251	109	9	1	9	75	268	478	660
	HDD18.3	6232	1009	811	761	499	322	137	78	142	302	526	728	918
	CDD10.0	550	0	0	0	2	49	137	203	136	23	1	0	0
	CDD18.3	48	0	0	0	0	3	14	22	9	0	0	0	0
	CDH23.3	537	0	0	0	0	50	183	226	78	1	0	0	0
	CDH26.7	118	0	0	0	0	10	54	44	10	0	0	0	0
	WSAvg	2.3	2.2	2.4	2.4	2.3	2.5	2.4	2.3	2.1	2.1	2.4	2.3	2.2
	PrecAvg	599	39	29	30	37	46	60	68	64	57	64	54	47
(15) Precipitation	PrecMax	754	77	62	69	69	99	125	126	154	118	114	105	102
	PrecMin	461	10	1	4	6	14	18	22	31	20	9	9	12
	PrecStd	67	14	14	15	17	19	25	28	26	23	24	22	20
	PrecStd	67	14	14	15	17	19	25	28	26	23	24	22	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	2.1	7.2	19.1	27.6	31.6	30.2	27.9	22.4	11.1	7.7	2.0
		MCWB	0.5	0.8	2.6	10.6	16.8	20.3	19.9	20.0	16.0	9.0	7.5	1.3
	2%	DB	-0.5	0.9	4.4	13.8	24.0	28.0	27.8	25.2	18.9	9.3	3.3	0.7
		MCWB	-0.7	0.2	1.0	8.1	14.4	18.5	18.6	18.5	13.6	8.1	3.0	0.1
	5%	DB	-2.6	0.2	2.8	10.8	21.3	25.1	25.8	23.2	16.7	8.2	1.9	0.2
		MCWB	-3.0	-0.4	0.3	5.6	12.8	16.7	17.6	17.2	12.7	7.3	1.5	-0.2
	10%	DB	-4.4	-1.1	1.6	8.5	18.5	22.7	24.1	21.0	14.5	7.0	1.1	-0.8
		MCWB	-4.9	-1.7	0.0	4.4	11.4	15.8	17.0	16.4	11.7	6.2	0.7	-1.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	1.2	3.2	11.3	18.1	21.8	22.2	21.1	16.6	9.8	7.3	1.4
		MCDB	0.6	1.6	6.2	18.0	25.9	27.8	27.0	25.8	19.9	10.7	7.7	1.7
	2%	WB	-0.8	0.4	1.7	8.5	15.8	19.9	20.0	19.5	14.7	8.4	3.1	0.3
		MCDB	-0.5	0.9	3.3	12.9	21.2	26.5	24.8	23.8	17.6	9.1	3.2	0.6
	5%	WB	-2.9	-0.4	0.9	6.3	13.7	18.5	18.7	18.3	13.3	7.4	1.6	-0.1
		MCDB	-2.5	0.2	2.1	9.4	19.4	23.5	23.6	21.9	16.0	8.1	1.8	0.1
	10%	WB	-4.8	-1.7	0.1	4.9	12.1	16.9	17.7	17.2	12.0	6.3	0.7	-1.2
		MCDB	-4.4	-1.2	1.4	7.8	17.5	21.3	22.1	20.0	14.2	7.0	1.0	-0.7
(35) Mean Daily Temperature Range	5% DB	MDBR	6.5	6.6	9.8	9.4	11.3	11.0	10.6	9.5	8.1	4.7	4.4	5.5
		MCDBR	5.7	4.1	10.4	13.0	15.9	15.1	14.0	12.5	13.4	5.9	3.3	3.6
		MCWBR	5.7	3.8	7.7	7.8	8.3	7.3	6.0	6.3	8.2	5.2	3.2	3.6
	5% WB	MCDBR	5.9	4.1	7.1	11.0	13.7	12.7	11.8	10.9	11.5	5.7	3.3	3.3
		MCWBR	5.9	3.9	5.4	7.0	7.7	6.9	6.1	6.0	7.6	5.3	3.3	3.4
(40) Clear-Sky Solar Irradiance	taub		0.249	0.296	0.328	0.359	0.356	0.357	0.398	0.374	0.338	0.314	0.260	0.227
	taud		1.980	2.052	2.064	2.163	2.431	2.463	2.351	2.398	2.483	2.365	2.075	1.934
	Ebn at Noon		394	631	756	819	864	868	815	805	772	637	384	229
	Edh at Noon		38	74	106	119	100	98	108	95	72	55	34	22
(44) All-Sky Solar Radiation	RadAvg		0.24	0.91	2.12	3.54	4.83	5.23	5.26	3.60	2.00	0.78	0.23	0.11
	RadStd		0.04	0.21	0.38	0.45	0.41	0.50	0.45	0.35	0.27	0.10	0.05	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	+0.60	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page